

Project Goal

Iterate on previous Comfort System designs to improve accuracy, reliability, and durability of the design while also facilitating rapid prototyping and testing for future improvements and iterations of the device.

Background

- Current assessment for long-term automotive seat comfort includes subjective human testing or non-representative manikin testing (**Figure 1**).
- 3 variables that are found to be most important in regards to seat comfort are pressure force, frictional force along the thigh, and vibration transmissibility.
- The existing prototype (**Figure 2**) incorporated some of these parameters, with vibration missing, by use of a variable thickness tissue mimicking layer, piezoresistive pressure sensors, and cylindrical piezoresistive stretch sensors.

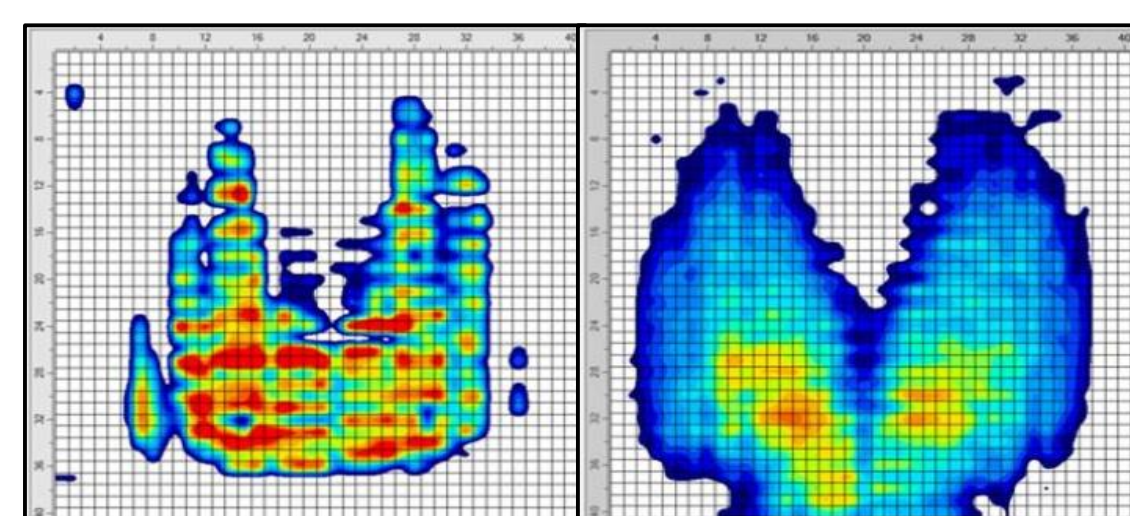


Figure 1: Manikin (left) seat pressure map versus human (right) seat pressure map

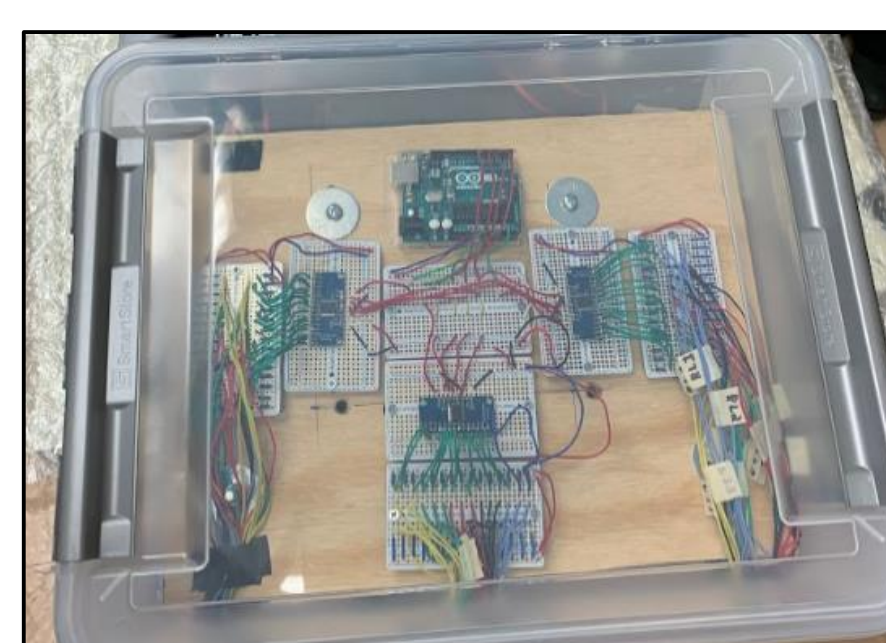


Figure 2: Lear Comfort System 2.0

Device Design

Software and Coding:

- The original Arduino code was translated to Python.
- The code was updated to record data faster, save the data to an on board SD card, and directly export as a CSV file for spreadsheet use.
- The original equations for all of the sensors were incorrect. The force sensors were coded to use a linear curve when they should have been using a logarithmic curve. The stretch sensors were using a force equation instead of the proper stretch formula. Once corrected, the sensors were re-calibrated accordingly or upgraded to have automatic calibration.
- Start and stop code was added for data collection, preventing the on-board microcontroller from corrupting when data collection was stopped during the recording of a data point.
- Accelerometer code was added to accomplish the vibration analysis.

In-House Testing:

- An in-house buttform was fabricated using 3D printing and a STL model provided by Lear.
- The buttform was weighed down using four 25lbs bags of lead (**Figure 3**) and was planned to be used for testing.



Figure 3: The assembled buttform loaded with lead weights.

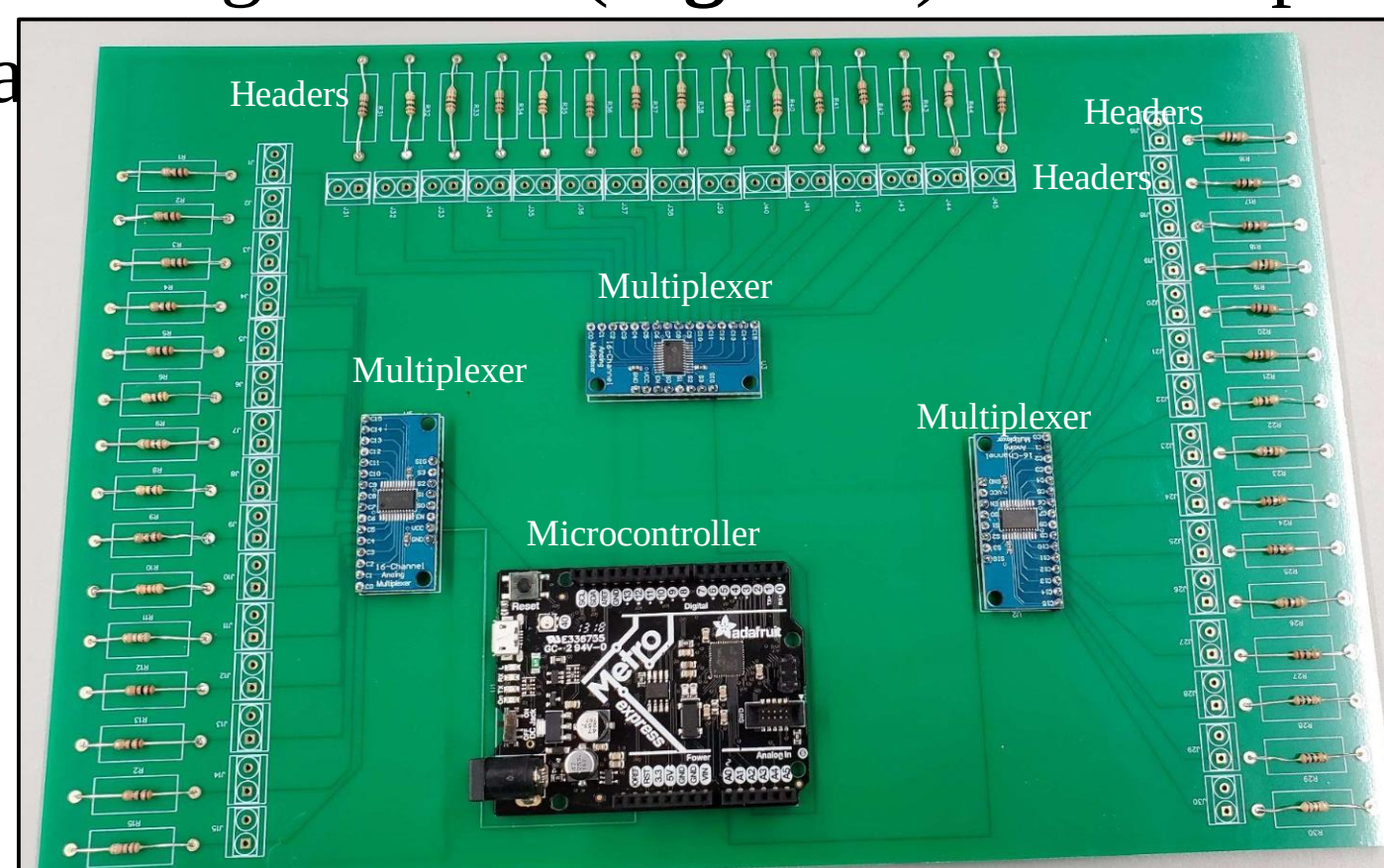


Figure 4: The PCB assembled using all components excluding sensor connectors.

Circuitry:

- The majority of wiring was accomplished by a printed circuit board (PCB) in the new iteration (**Figure 4**). The PCB included the voltage dividers, the microcontroller, the multiplexers, and female headers for quick release of the sensor wiring (see **Figure 4** for labeling of the components).
- Sensor attachments were rebuilt into the final prototype wiring to allow for simplified reconfiguration.

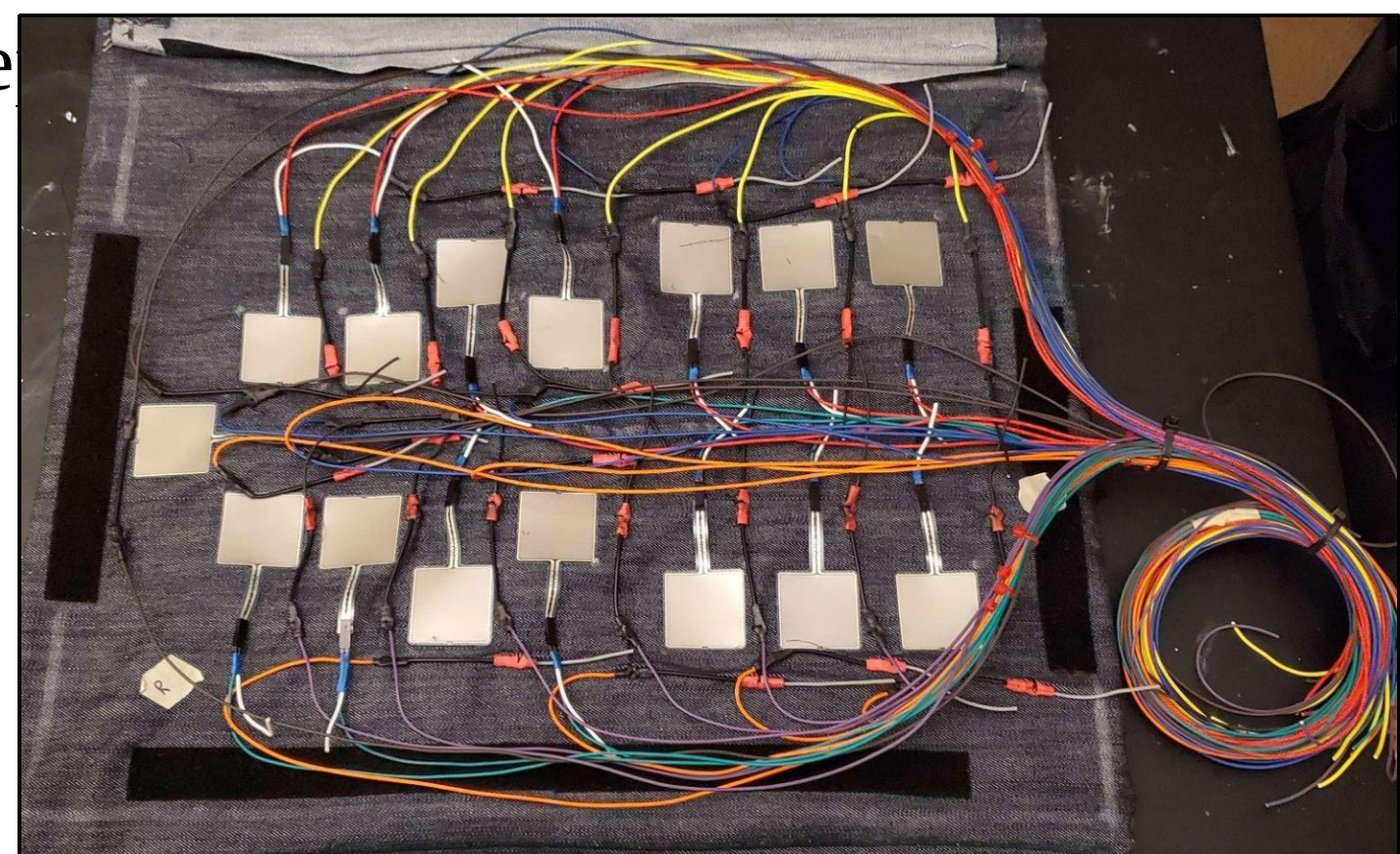


Figure 5: The completed device wiring for the sensors on the denim fabric.

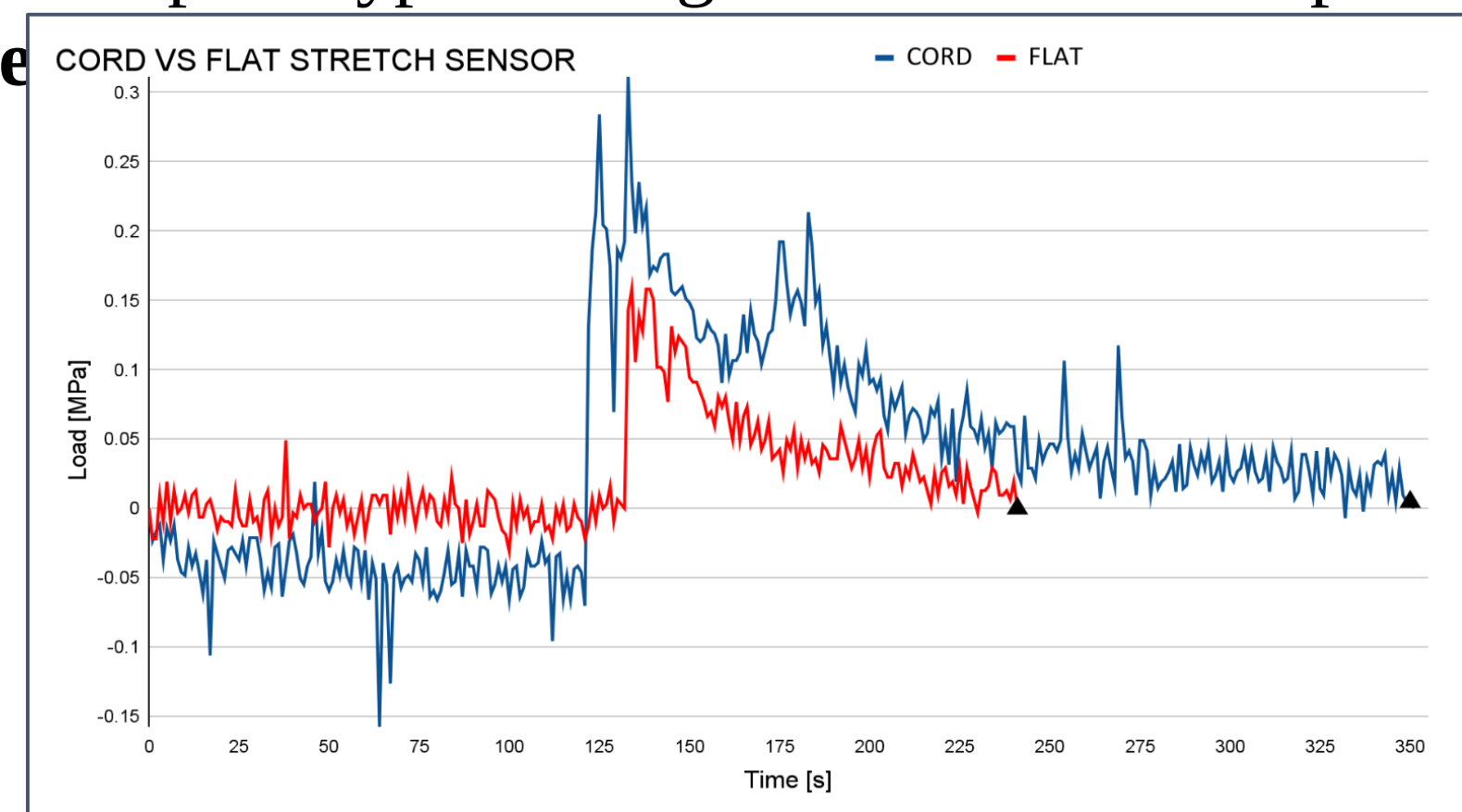


Figure 6: Stretch sensor hysteresis graph. Black triangles were included to mark when the curve returned to zero.

Prototyping:

- The existing prototype was rebuilt to improve the durability of the design. 30 AWG silicon wire was replaced with 20 AWG TXL wire rated for automotive use. The polyester fabric used was replaced with denim to create a more reliable and realistic design.
- Sewn-in-place stretch sensors with eyelets were replaced by plastic clamps to allow easy tensioning of the sensors. These were also changed from cylindrical sensors to flat sensors. This change made it so that the stretch sensors would not be affected by vertical loading to the same degree, collecting loads parallel with the surface of the device instead. This also served to minimize issues with the cylindrical sensors wherein the sensors had a lengthy recovery time, a hysteresis effect, before returning to an appropriate zero point (**Figure 6**). The hysteresis effect reduced the efficiency and accuracy of readings from the design.

Sensor Calibration and Validation

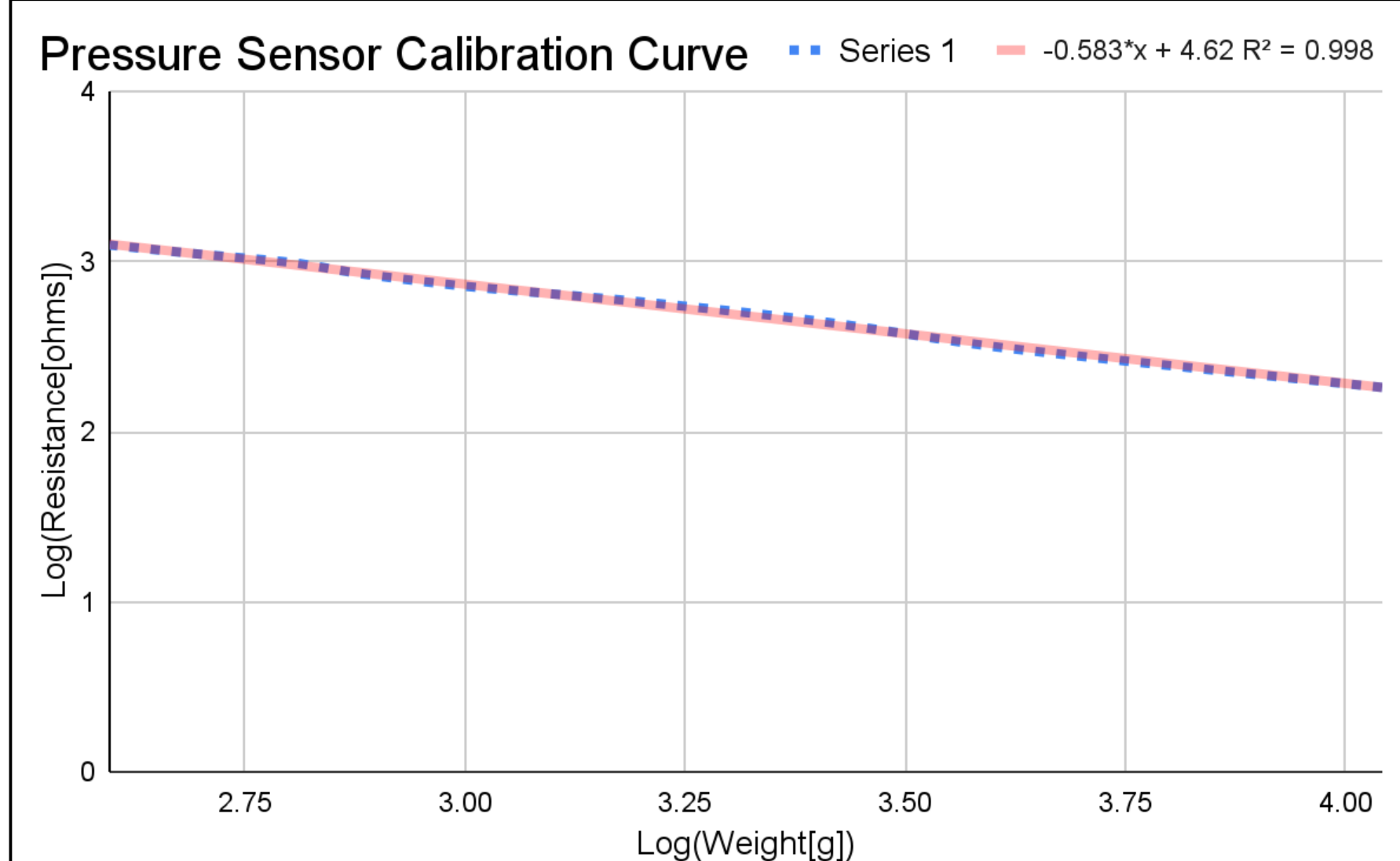


Figure 7 (Left): Pressure sensor calibration. This curve represents the logarithmic relationship between resistance and applied load to the pressure sensors utilized for this system. This relationship was utilized to create an equation to reliably and reproducibly determine the load applied to the sensor in a given instance. Every one of the 15 pressure sensors were calibrated by hand and calibration values were integrated into the finalized software after verifying the accuracy of each sensor independently.

Figure 8 (Right): Stretch sensor material testing.

Both the original sensors and the replacement sensors utilized a piezoelectric response to determine applied load. Electrical resistance in the sensor could be used to determine deformation, by multiplying the deformation by the elastic modulus taken from the slope of the curve above it was possible to calculate the applied load. The curve above represents the stress vs strain response of the aforementioned sensor material. Based on this, it was possible to incorporate autocalibration of these sensors on startup in the device software by treating the initial resistance as a zero point and using increases in resistance to calculate deformation.

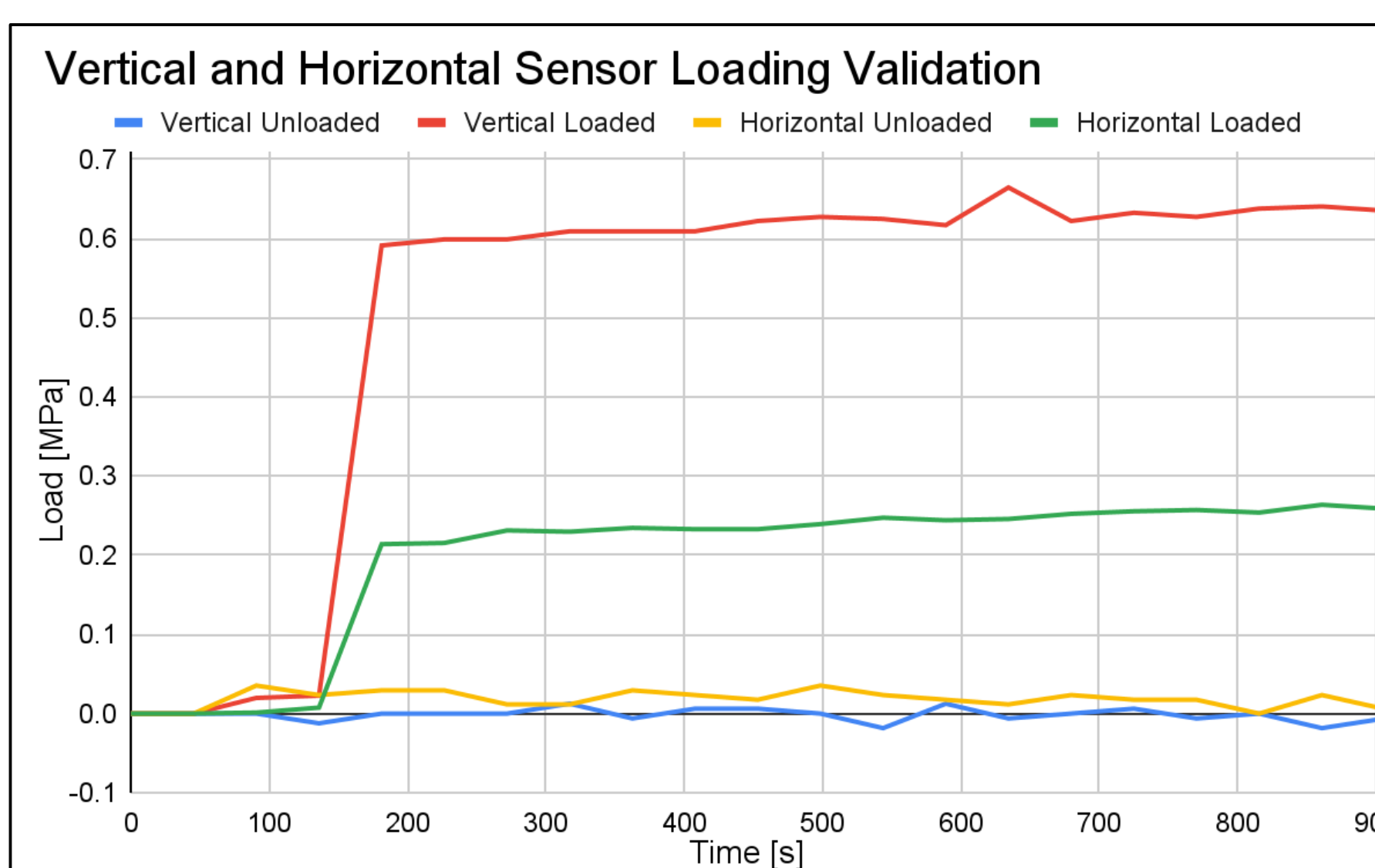
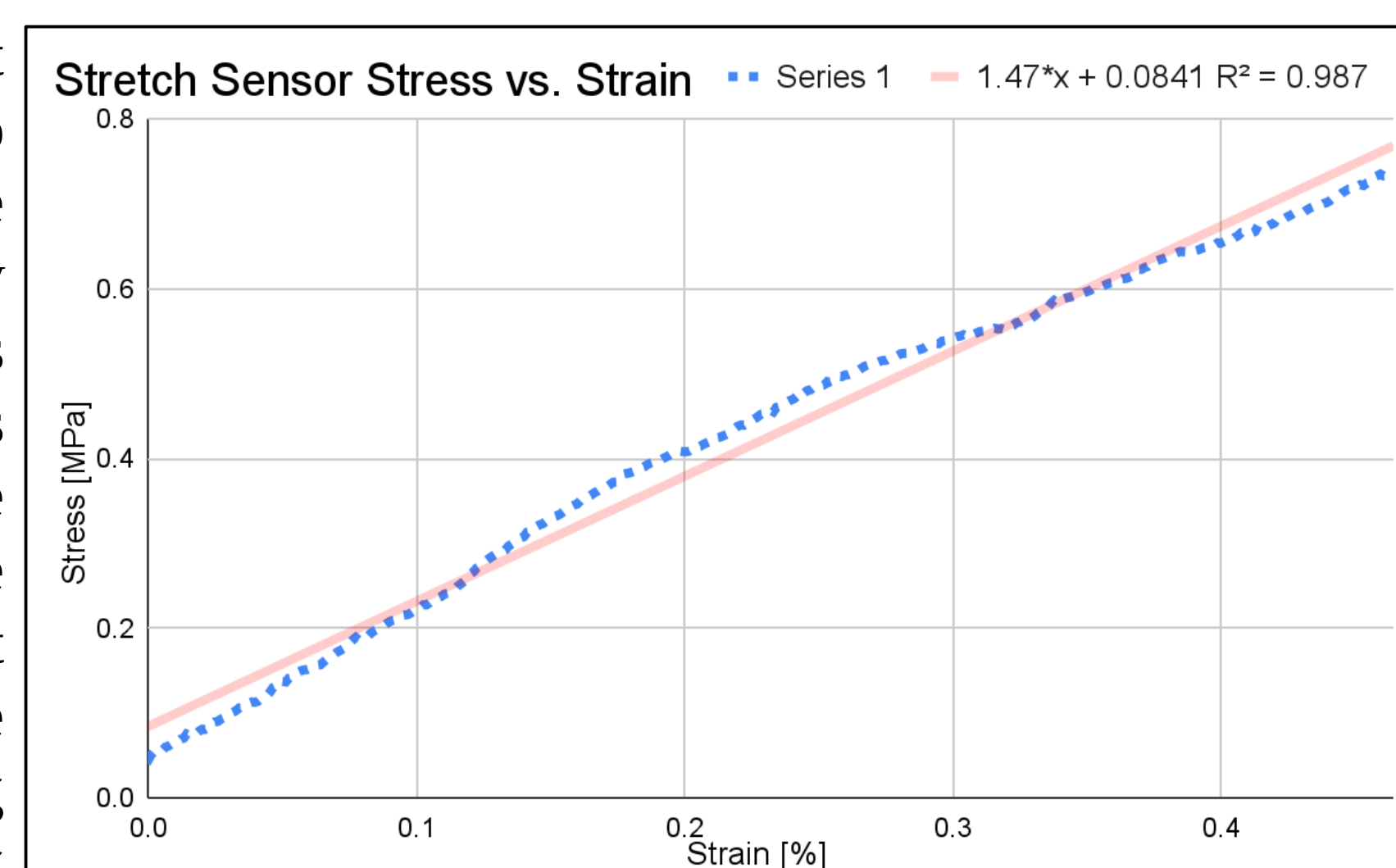


Figure 9 (Left): Graph of the vertical and horizontal loading validation of a single sensor node. 100lbs of lead was loaded onto the device which resulted in the changes above. Of the weight loaded, about 25% was distributed into the horizontal direction and 75% was distributed in the vertical direction. The loads applied on both sensors were able to be translated back to a force given the area of the sensors. The summed forces were very close to 100lbs. This proved the accuracy in the developed stretch sensors.

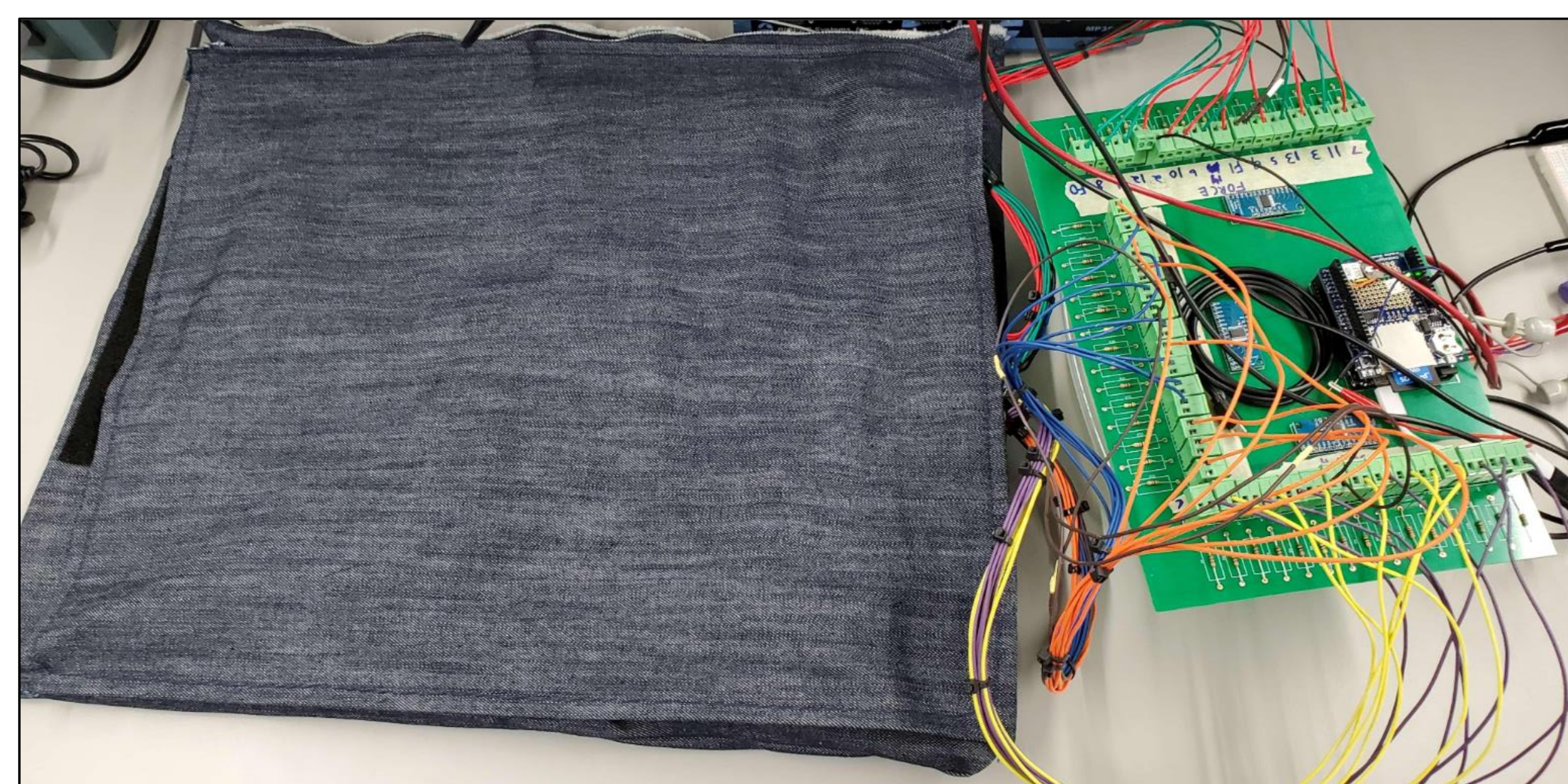


Figure 10: Full scale Lear Comfort System prototype. This design features two parts, the sensor array and the printed circuit board (PCB). The sensor array consists of the 45 sensors with 30 of them being upgraded from cylindrical stretch sensors from previous years to flat stretch sensors to more accurately record the proper data. This also overcomes issues in prior years testing which resulted in long periods of recovery before stretch sensors would return to zero. The PCB houses the 3 multiplexers that each connect to 15 sensors and make it possible to read all 45 sensors with limited microcontroller inputs. The PCB also houses the removable headers for the sensor wiring as well as the microcontroller itself.

Limitations, Conclusions, and Future Work

Limitations

- Due to the poor quality of the microplexers, there was a significant amount of aliasing, so much so that sensor data was a sum of multiple sensor inputs. This resulting is data collection being limited to one data point every 45 seconds for a given sensor
- The microcontroller internal storage was being pushed to the max, which limits the possibility to add more components to the design

Conclusions

- The new code for the sensors corrected the incorrect equations to give correct data readings
- The denim material improved device durability and realism
- The flat stretch sensors reduced noise in the readings and improved the recovery time

Future Work

- Replace the three multiplexers with a single 64 channel data acquisition system (DAQ)
- Derive a model for outputting a comfort score using the sensor readings
- Validate the device's performance using human subject testing

Acknowledgements

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